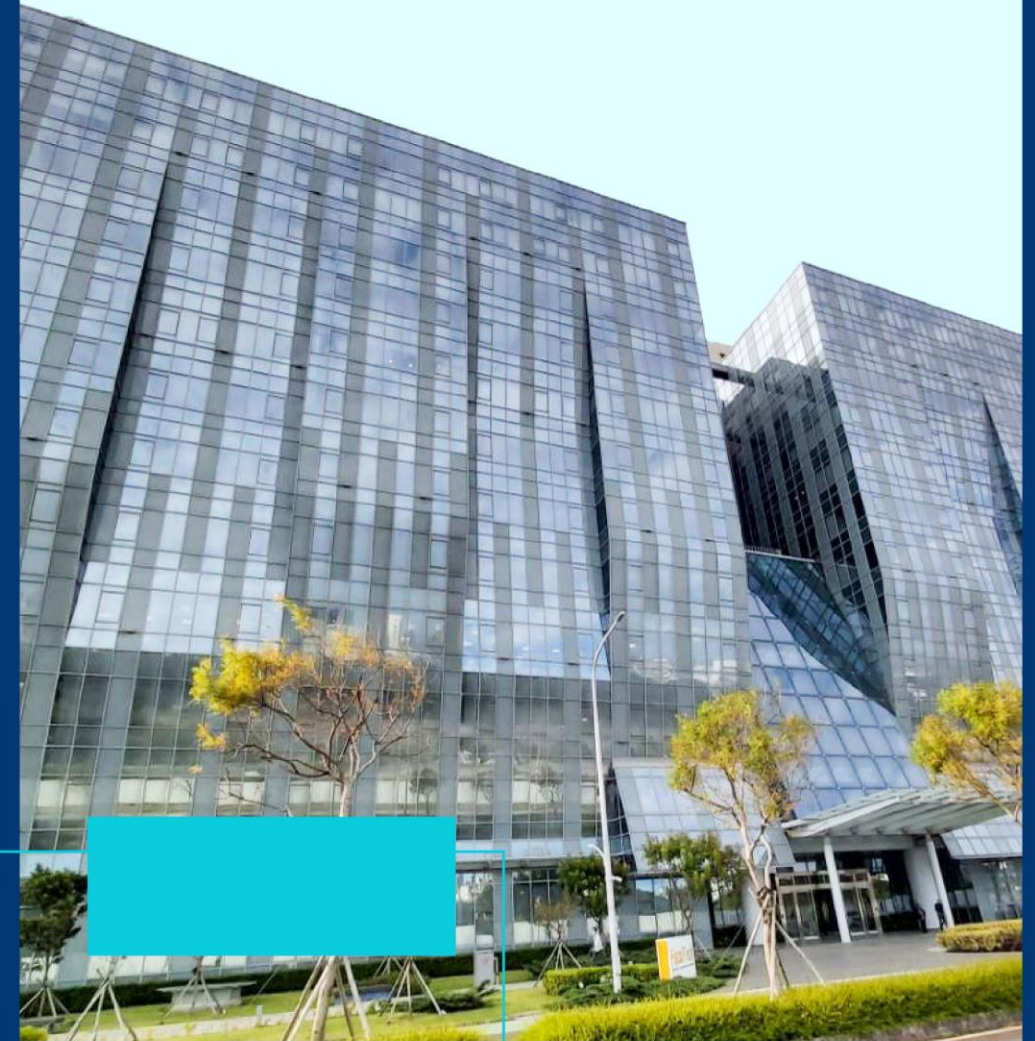
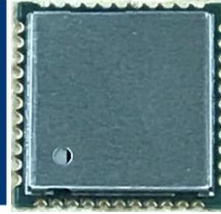


Advanced Module Packaging Solution

2026

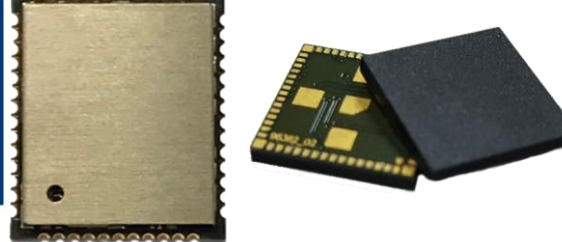


1-Stamp Type(1T1R)



Ant. Diversity Low Power High Power 3 Antennas GNSS Wi-Fi /BT 2.4/5/6 Ghz						
Model Name	Size	Wi-Fi	BT/802.15.4	Bandwidth	Interface	Status
✓ AP6256	12*12*1.5mm	Wi-Fi 5 1x1 2.4G/5G	5.4 Dual-Mode	80Mhz	SDIO3.0 + UART	Production
✓ AP5256/AP5856	8*8*1.4/1.2mm	Wi-Fi 5 1x1 2.4G/5G	5.4 Dual-Mode	80Mhz	SDIO3.0 + UART	Production
✓ AP6611S	12*12*1.5mm	Wi-Fi 6E 1x1 2.4G/5G/6G	5.3 Dual-Mode /LE Audio	80Mhz	SDIO3.0 + UART	Production
✓ AP5811SA	7.3*7.9*1.1mm	Wi-Fi 6E 1x1 2.4G/5G/6G	5.3 Dual-Mode /LE Audio	80Mhz	SDIO3.0 + UART	Production
✓ AP6281S	12*12*1.75mm	Wi-Fi 6E 1x1 2.4G/5G/6G	5.3 Dual-Mode /LE Audio /802.15.4	80Mhz	SDIO3.0 + UART	Production
✓ AP5281SR	6.4*7.9*1.1mm	Wi-Fi 6E 1x1 2.4G/5G/6G	5.3 Dual-Mode /LE Audio /802.15.4	80Mhz	SDIO3.0 + UART	Production

2-Stamp Type(2T2R)



Ant. Diversity

Low Power

High Power

3 Antennas

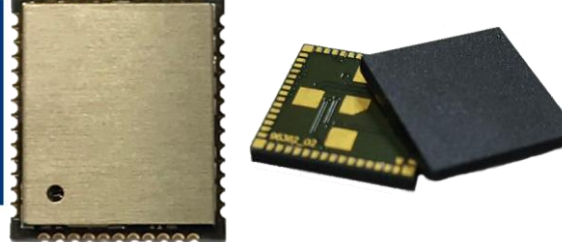
GNSS

Wi-Fi /BT

2.4/5/6Ghz

Model Name	Size	Wi-Fi	BT/802.15.4	Bandwidth	Interface	Status
✓ AP6676P	13*15*1.65mm	Wi-Fi 6E 2x2 2.4G/5G/6G	5.3 Dual-Mode/ LE audio	80Mhz	PCIe + UART	Production
✓ AP5876EPA	10*10*1.12mm	Wi-Fi 6E 2x2 2.4G/5G/6G	5.3 Dual Mode/ LE Audio	80Mhz	PCIe+ UART	Production
✓ AP56676S	13*15*1.65mm	Wi-Fi 6E 2x2 2.4G/5G/6G	5.3 Dual Mode/ LE Audio	80Mhz	SDIO+ UART	Production
✓ AP5876SE	10*10*1.12mm	Wi-Fi 6E 2x2 2.4G/5G/6G	5.3 Dual Mode/ LE Audio	80Mhz	SDIO+ UART	Production
✓ AP6275P	13*15*1.65mm	Wi-Fi 6E 2x2 2.4G/5G	5.3 Dual-Mode/ LE audio	80Mhz	PCIe + UART	Production
✓ AP5875EPA	10*10*1.12mm	Wi-Fi 6E 2x2 2.4G/5G	5.3 Dual Mode/ LE Audio	80Mhz	PCIe + UART	Production

2-Stamp Type(2T2R)



Ant. Diversity

Low Power

High Power

3 Antennas

GNSS

Wi-Fi /BT

2.4/5/6Ghz

Model Name	Size	Wi-Fi	BT/802.15.4	Bandwidth	Interface	Status
✓ AP6275HH3	24*24*1.6mm	Wi-Fi 6 2x2+2x2 RSDB 2.4G/5G	5.3 Dual Mode/ LE Audio	80Mhz	PCIe + UART	Production
✓ AP6684P Syna4384_28nm	13*15*1.6mm	Wi-Fi 7 1x1+1x1 RSDB 2.4G/5G/6G	6.0 Dual-Mode_ LE audio/802.15.4	80Mhz	PCIe + UART	Production
AP5684P Syna4384_28nm	10*10*1.15mm	Wi-Fi 7 1x1+1x1 RSDB 2.4G/5G/6G	6.0 Dual-Mode_ LE audio/802.15.4	80Mhz	PCIe + UART	2026/Q4 sample
AP6690P BRM4390_6nm	17*17*1.95mm	Wi-Fi 7 2x2+2x2 RSDB 2.4G/5G/6G	6.0 Dual-Mode_high power/ LE audio/802.15.4	160Mhz	PCIe + UART	Sample ready
AP5890P BRM4390_6nm	13*15*1.2mm	Wi-Fi 7 2x2+2x2 RSDB 2.4G/5G/6G	6.0 Dual-Mode_high power/ LE audio/802.15.4	160Mhz	PCIe + UART	2026/Q4 Sample

Power Consumption Comparison

Soft AP mode	5G TX Tupt(380~400Mbps)		5G RX Tput(390~420Mbps)		2G TX Tput(100~110Mbps)		2G RX Tput(105~120Mbps)	
Module name	3.3V_mA	1.8V_mA	3.3V_mA	1.8V_mA	3.3V_mA	1.8V_mA	3.3V_mA	1.8V_mA
AP5281SR	167	2.57	76	2.17	43.66	1.93	53	2.79
AP5811SA	324	4.5	136	2.22	83.2	2.19	79	2.16
AP5256/AP5856	280.73	2.45	173	6.04	92.1	2.11	80.6	5.93
	Tput >800Mbps		Tput >800Mbps				Tput>220Mbps	
AP5875EPA	630		253	2.45			128	1.89
AP5876EPA_5G	420		206	1.8			133	1.7
AP5876EPA_6G	425.9							

Module name	BLE connection(1s interval)		BLE unconnectable adv(1s interval)	
	VBAT	VDDIO	VBAT	VDDIO
AP5281SR	0.404(3.3v)	0.068(1.8v)	0.384(3.3v)	0.068(1.8v)
AP5811SA	0.133(3.3v)	0.1(1.8v)	0.18(3.3v)	0.06(1.8v)
AP5256	0.062(3.6v)	0.13(1.8v)	0.073(3.6v)	0.131(1.8v)
AP5875EPA	0.148(3.3v)	0.11(1.8v)	0.184(3.3v)	0.11(1.8v)
AP5876EPA	0.136(3.3v)	0.11(1.8v)	0.152(3.3v)	0.11(1.8v)

Power Consumption Comparison

STA mode	AP5281SR_5dBm	<u>AP5281SR_11.5dBm</u>	<u>AP5281SR_12.25dBm</u>	<u>AP5281SR_9.5dBm</u>	<u>AP5281SR_8.25dBm</u>
5G Tput=100Mbps		138	137.6	93	85
5G Tput=40Mbps	44.9	94.2	81.4	66.8	59.7
5G Tput=20Mbps	24.5	63	60	53.8	50
5G Tput=15Mbps	18	58	56.3	50.3	46.7
5G Tput=10Mbps	12.9	50.9	50	48.4	44.1
5G Tput=5Mbps		43.2	41.1	40.5	38.1
5G Tput=1Mbps		20.3	19.1	20.6	17.1

Border Router Applications

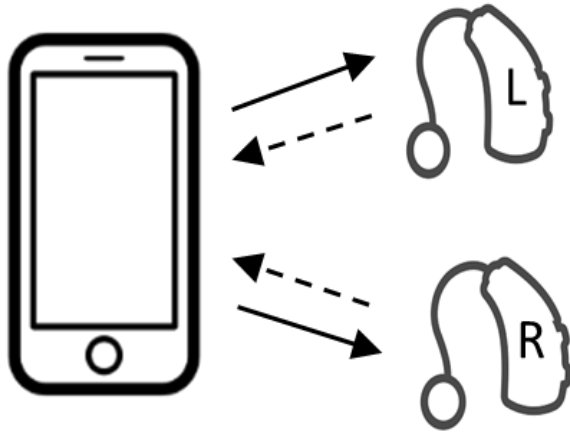
- AP6281S/AP6612X support Wi-Fi, Bluetooth and 802.15.4(Thread) that can be used as Border Router.



LE Audio Support

- Full LE audio feature
 - LE Isochronous Channels support

LE Audio Stereo(CIS)
SRC/SNK



LE Audio Auracast(BIS)
SRC/SNK



Bluetooth channel sounding

Channel Sounding (aka HADM) vs. Ultra Wide Band (UWB)

	BT CS	UWB
Accuracy	30~50 cm	5~30cm
Distance	Covers long range	Optimal 1 to 30m
System Cost	More circuit in IC. No additional System cost	UWB SoC Cost: \$4+ 3 antennas - (2 for RX, 1 for TX) Extra PCB Space
Power consumption	Low : 2.4GHz BT carrier	Very high : 3 to 10GHz operation
Core Tech	Carrier Signal Phase	Time Measure (radar)

- Both technologies use time and frequency domain
- The accuracy varies depending on the channel environment

Bluetooth Channel Sounding enabled on 4 or 6 anchors



Smart car key
Keyless auto-lock



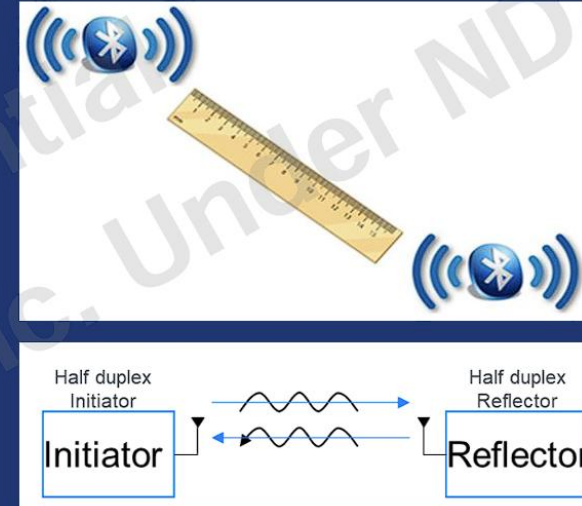
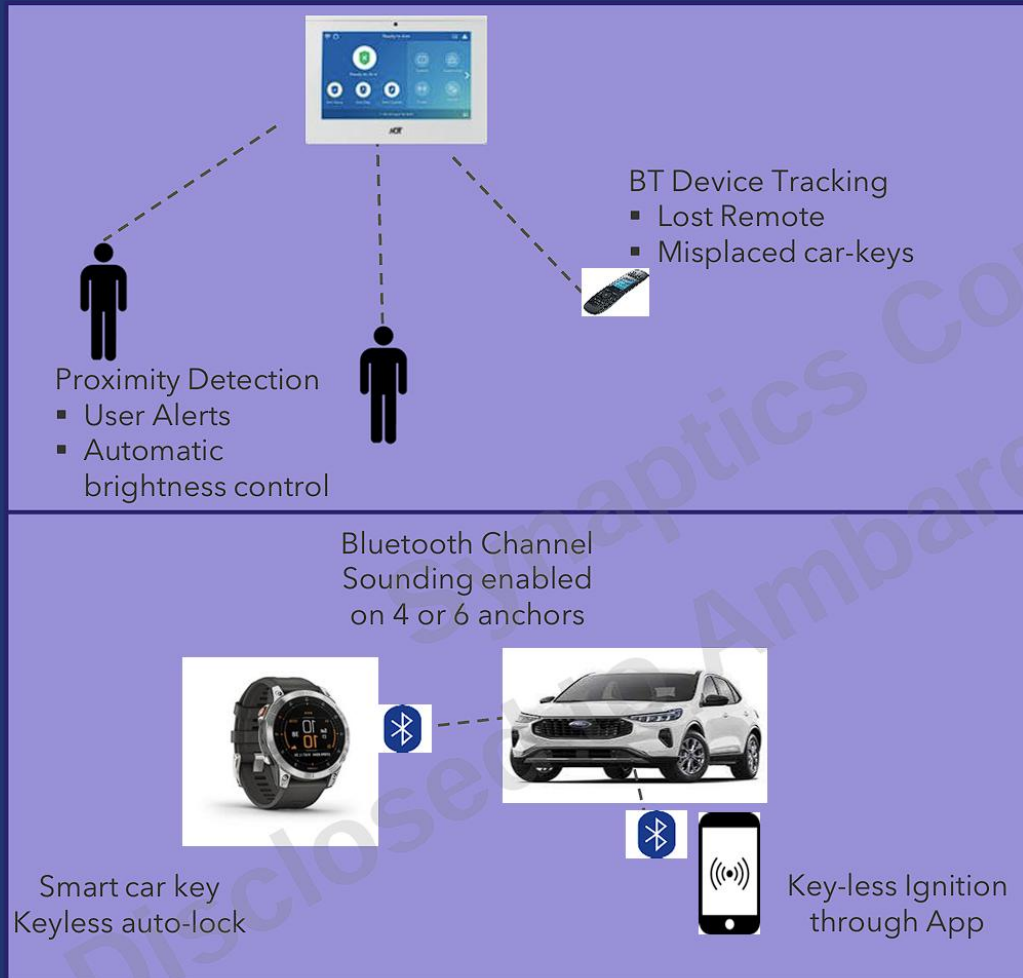
Key-less Ignition through App



4382

Bluetooth channel sounding

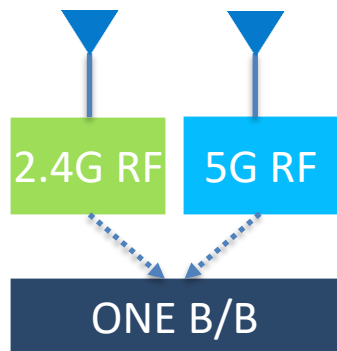
BLUETOOTH CHANNEL SOUNDING



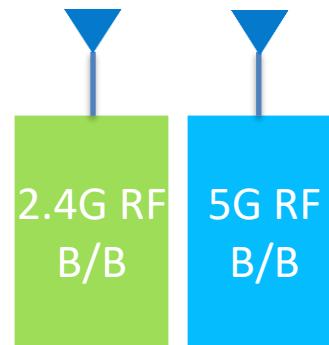
- Channel Sounding – High Accuracy Distance Measurement method between two Bluetooth devices
- High Accuracy Distance Targets
 - Up to 5m with an accuracy of $\pm 0.5\text{m}$
 - Above 5m with an accuracy of $\pm 10\%$

RSDB Feature

- AP6398S, AP6398P, AP6275HH3, AP6682SR3, and AP6684P supports the RSDB feature to fulfill concurrent user scenarios.



- VSDB:
Two separate RFIC but
one baseband requires a
digital switching



- RSDB:
Two separate RFIC
and baseband cores

2.4G Traffic flow



5G Traffic flow



2.4G Traffic flow

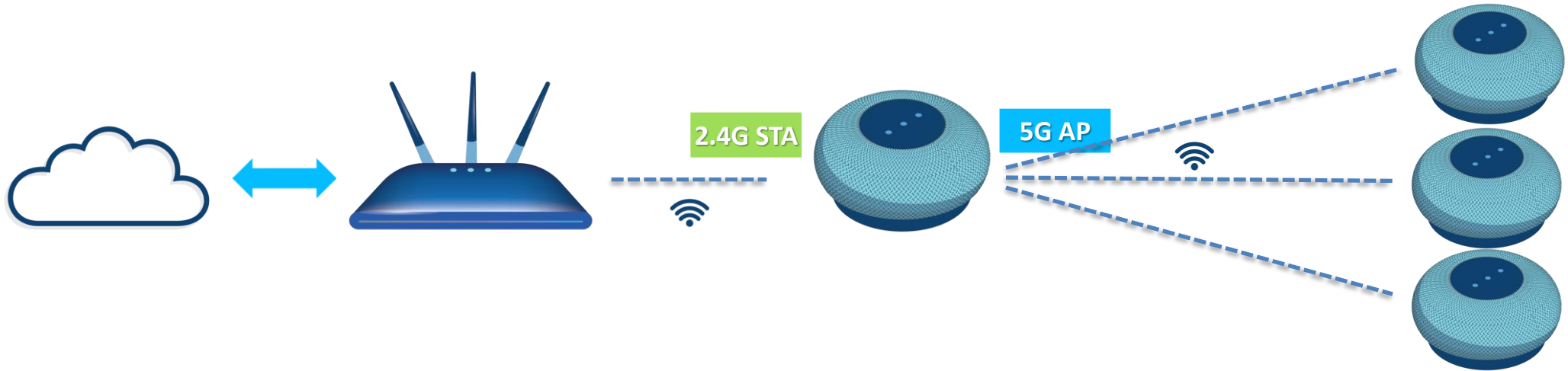


5G Traffic flow



RSDB User Scenarios

- Case 1: Station/SoftAP concurrent



- RSDB advantages:
 - No delay
 - No throughput degradation

AMPAK is your best partner



Thank You

Advanced Module Packaging Solution